Mithi River Hybrid Model Accuracy Report

Architecture: CEEMDAN-Attention-BiLSTM-XGBoost

Decomposition: moving_average_residual_fallback

Training Configuration

Window size	6
Sample rows	20,000
Epochs	8
Batch size	128
Learning rate	0.001
Selected features	5

Classification Accuracy

Metric	Result
Accuracy	96.90%
Macro F1	0.3281
Weighted F1	0.9537
Test support	2,996

Per-Class Classification Results

Class	Precision	Recall	F1	Support
Good	0.0000	0.0000	0.0000	5
Moderate	0.0000	0.0000	0.0000	88
Poor	0.9690	1.0000	0.9842	2903

Regression Performance: TDS, BOD, and COD

Metric	Result
Mean absolute error (MAE)	424.29
Root mean square error (RMSE)	817.89
R^2 score	-0.0103

Interpretation

The model reports 96.90% classification accuracy, but this headline number is not sufficient for deployment because the dataset is highly imbalanced. The model correctly identifies the majority Poor class while showing zero recall for Good and Moderate in this test split. Macro-F1 is therefore the more important classification metric and currently indicates weak minority-class performance. The negative regression R^2 indicates that the regression component requires further improvement before production use.

Selected classification features: COD, BOD, DO, pH, TDS

Selected regression features: TDS, COD, BOD, Temp, pH